



THE MEDIA CUSTOMS

Operator manual



Google



What is in this manual

It is written to be read in order once, then opened at the chapter you need. Every screen it describes is live, and every number in it was read from the running system rather than typed from memory.

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What this is

An agentic ad-clearance crew. It watches a commercial once, judges it against every market you ship to in parallel, and re-renders the shots that fail.

A commercial is cut once and shipped everywhere. Somebody then has to answer, per market, whether it can legally air. Today that answer comes from a lawyer for the two or three biggest markets and from hope for the rest, which is how a gesture that is friendly here becomes an insult there, and how an ad gets pulled after it airs rather than before.

The Media Customs answers it for all of them at once, and shows its working. Every finding names the shot, the seconds, the rule, and the statute behind the rule, with a citation resolved live at the moment of judging. Nothing in the console is a summary of something you cannot open.

Who operates it

ROLE	WHAT THEY DO HERE
Producer or traffic	Starts the clearance, reads the board, exports the certificate and the markers.
Legal or compliance	Reads the findings and their statutes, decides the ones the Guard hands over.
Editor	Takes the markers into Resolve or Premiere, or reviews what the remediator rendered.

What it produces

- A decision per market, cleared, at risk or blocked, with the evidence behind each.
- A localized master per market, with only the failing seconds touched.
- A clearance certificate as a PDF, every finding and statute on it.
- Marker files for an editing suite, colour-coded by what the finding does.

One sentence to remember

A finding is always a join: **this observation, times that market's rule, times a citation that resolves**. Break any leg of it and the finding is not made. That is why you can argue with a finding usefully, and why the system can tell you which leg is wrong.

How it thinks

Looking at the film is the expensive part. Opinions about what was seen are cheap. So the two never mix.

One multimodal analyst watches each shot a single time and writes down what it sees, in neutral language, with a timecode and a box. It is forbidden an opinion. It writes *a woman raises a glass of red wine*, never *this violates French law*.

That single fact sheet then goes to every selected market at once. One adjudicator per jurisdiction reads it against its own rulebook and grounds each citation against a live source. Ninety-eight adjudicators argue about the same sentence, simultaneously, each holding a different rulebook.

Why it is built this way

- **It is cheap.** Adding a market costs one more text call, not one more viewing of the film.
- **It is parallel.** Markets do not queue behind each other.
- **It is arguable.** "Is this fact wrong?" and "is this rule wrong?" become separate questions with separate answers, which is what makes a disagreement resolvable.

The vocabulary is fixed

The analyst may only emit observations under eighteen dimensions, and every one of the 128 rules is written against one of them. That is what makes the join between a fact and a market's opinion a lookup rather than an argument.

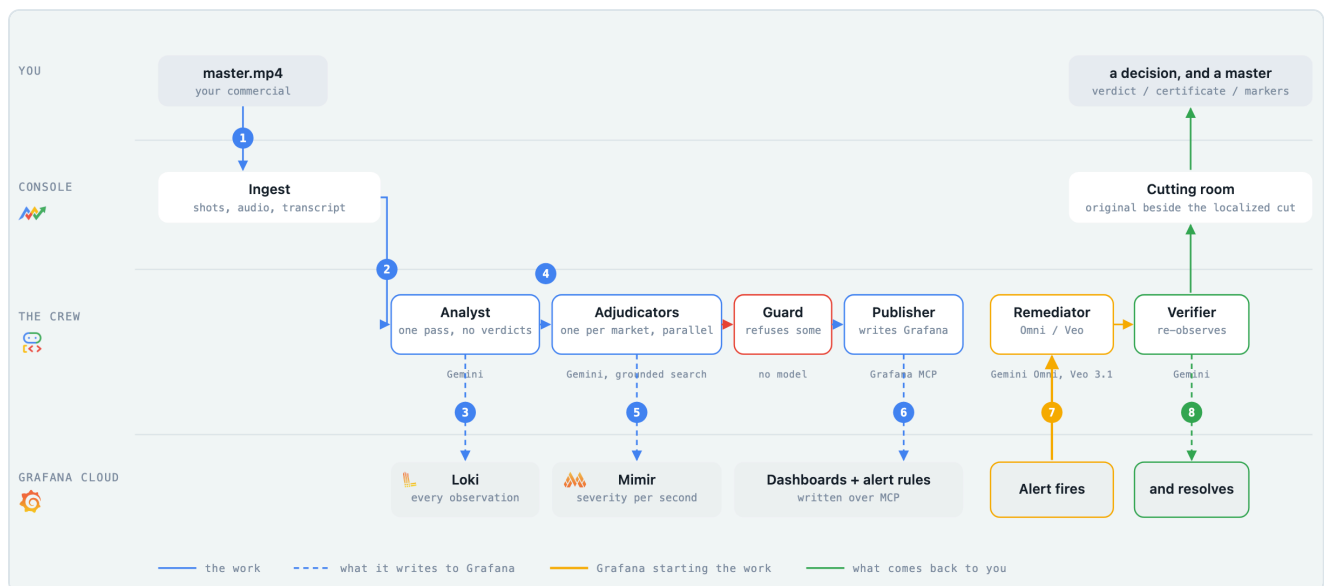
alcohol, tobacco, drugs	religious symbols	modesty and dress
gesture and body language	food and animals	gender portrayal
orientation and gender id	children and minors	national symbols
health claims and pharma	gambling and finance	violence and weapons
language and idiom	humour and satire	superstition and colour
photosensitivity	on-screen text	comparative claims

The ladder

Markets inherit. A run against a Belgian broadcaster is judged against that channel's own acceptance rules, Belgian law, the EU directive and the global baseline, in that order. Twenty-one packs on disk resolve into 98 selectable jurisdictions.

A run, end to end

Eight stages. Each one writes something you can open afterwards.



The same diagram is on the console's front page, and it plays itself. Every stage names the model doing its job underneath it.

How long it takes

About four minutes for a thirty second spot, and proportionally longer for longer films, because the analyst reads every shot: shot detection, a transcription, an observation pass per shot, then every market judged in parallel with a grounded citation per finding.

The arrow that matters

Stage 6 is the one that makes this different from a report. Nothing polls, and no cron runs. A threshold crosses on a dashboard the crew built itself, Grafana calls the service, and generation begins. The fix starts because a chart said so.

The eight stages

What each one does, and what it leaves behind for you to open.

#	STAGE	WHAT HAPPENS	WHAT IT LEAVES BEHIND
1	You	Hand it a master and tick the markets.	The run, in the archive.
2	Ingest	Cuts the film into shots, pulls the audio and a transcript.	Shots and keyframes.
3	Analyst	Watches each shot once. Neutral, timecoded, boxed observations.	Loki lines, kind=observation .
4	Adjudicators	One per market, in parallel, each against its own rulebook, each citation grounded.	Findings, with statutes.
5	Publisher	Pushes metrics and lines, and builds the run's dashboards and alert rules.	Mimir series, annotations, 9 boards.
6	Alert	A blocking finding trips a rule Grafana is already evaluating.	A webhook call into the service.
7	Remediator	Plans, prices and edits the failing span. One edit per shot.	A change record and a staged file.
8	Verifier	Re-runs the real analyst on the changed shots.	Fixed, or the finding back to open.

Where each stage shows up in the console

Stages 2 to 4 are the mission feed and the frame board. Stage 5 is the launch board and everything on the Grafana page. Stages 6 to 8 are the market room, the fix panel and the cutting room. Nothing happens that you cannot afterwards open and read.

Before you start

What you need, what it costs, and the things it will refuse to do.

Access

Reading is never gated. The archive, every run, every finding with its statute and citation, the rule library, frame search, the intelligence board and the Grafana inventory are open to anyone with the link. What is gated is everything that **spends or destroys**: starting a clearance, judging more markets, running a fix, asking the agent, and deleting a change record. Those ask for a word once and remember it in a cookie.

Two doors

The visitor door lifts a daily spending cap of one euro. The judge door lifts it entirely. Having the word is not being someone: it is a speed bump, and the ceiling behind it is what actually bounds a stranger's spend.

What a run needs

INPUT	ACCEPTED	LIMIT
A master	MP4 or MOV	Up to 128 seconds, up to 30 MB by upload
A link	Public video URL	Same duration cap, fetched at up to 720p
Markets	Any of 98	More markets is more cost and more wall clock

What it will not do

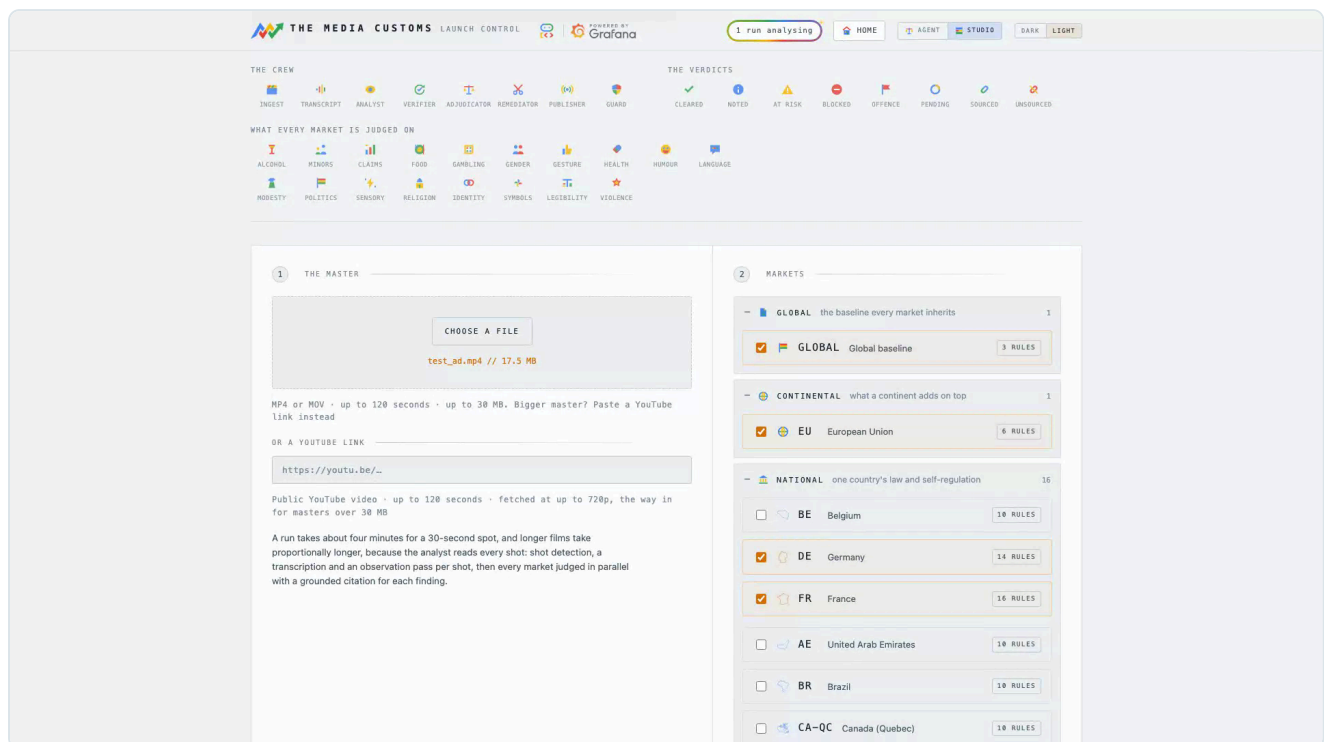
- **It will not auto-edit a protected characteristic.** Two rules in the corpus are written on one, and a finding matching either is handed to a person instead. Chapter 14.
- **It will not spend past the day.** A budget alert pauses automatic remediation when what is left drops below the price of the most expensive single fix. Chapter 22.
- **It will not keep an edit that damaged the film.** The craft gate discards it and the finding goes back to open. Chapter 12.
- **It will not trust a model's word that a fix worked.** The verifier re-observes.

Read this before you upload borrowed footage

Gemini Omni refuses to edit footage containing recognisable third-party content, and Veo refuses a shot it reads as depicting a public figure. Neither is charged, and a retry cannot help: the footage is the refusal. Patch methods still work on that material, but the archive is public to anyone with the link, so only upload what you have the rights to show.

Starting a clearance

The launcher asks two things: the film, and who has to accept it.



The master on the left, the market ladder on the right.

Step by step

1. **Choose the master.** Drop a file or paste a link. The duration is checked before anything is downloaded, so an over-long film is refused without spending.
2. **Open the ladder and tick markets.** Global and the EU are ticked by default. Countries sit under the continental rung, broadcasters under their country.
3. **Read the price,** shown against the day's remaining budget.
4. **Begin clearance.** The console moves to the mission feed and the crew takes over.

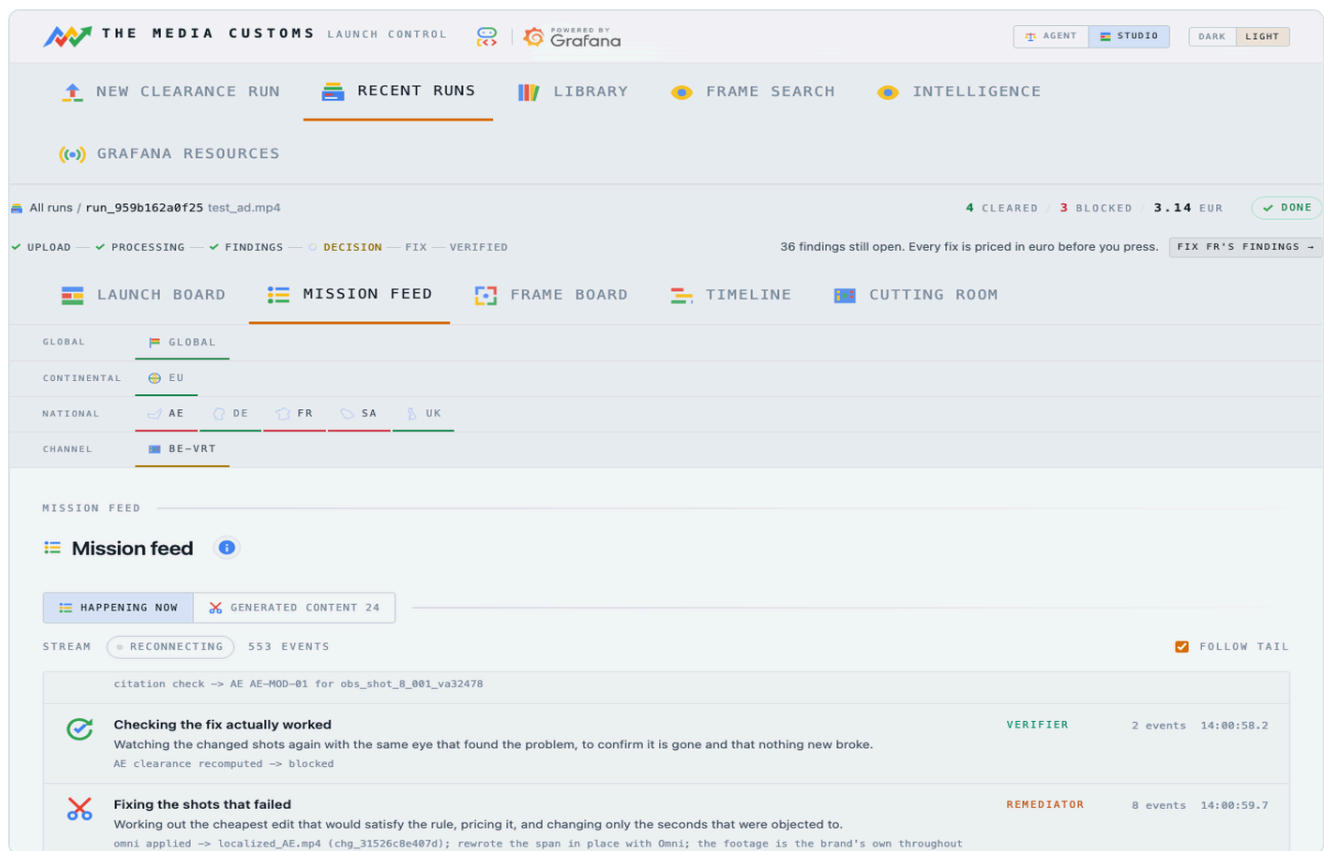
Choosing markets well

IF YOU WANT	TICK
A fast sanity check	Global and the EU: two adjudicators, the broadest rules, about a minute.
A real clearance	Every country you will air in. National law is where the blocking findings live.
A channel delivery	The broadcaster itself. It inherits its country and adds its own rules.

A run is cheap; a fix is not. Judging costs cents and generation costs euro, so tick generously here and be deliberate later.

Watching it work

The mission feed is the crew's own event log, streamed as it happens.



Every move, in the order it was made. Not a spinner.

Each stage says what it is doing in plain words, with its own shorthand underneath. A stage that fails says so here rather than skipping quietly, and counts itself as a metric.

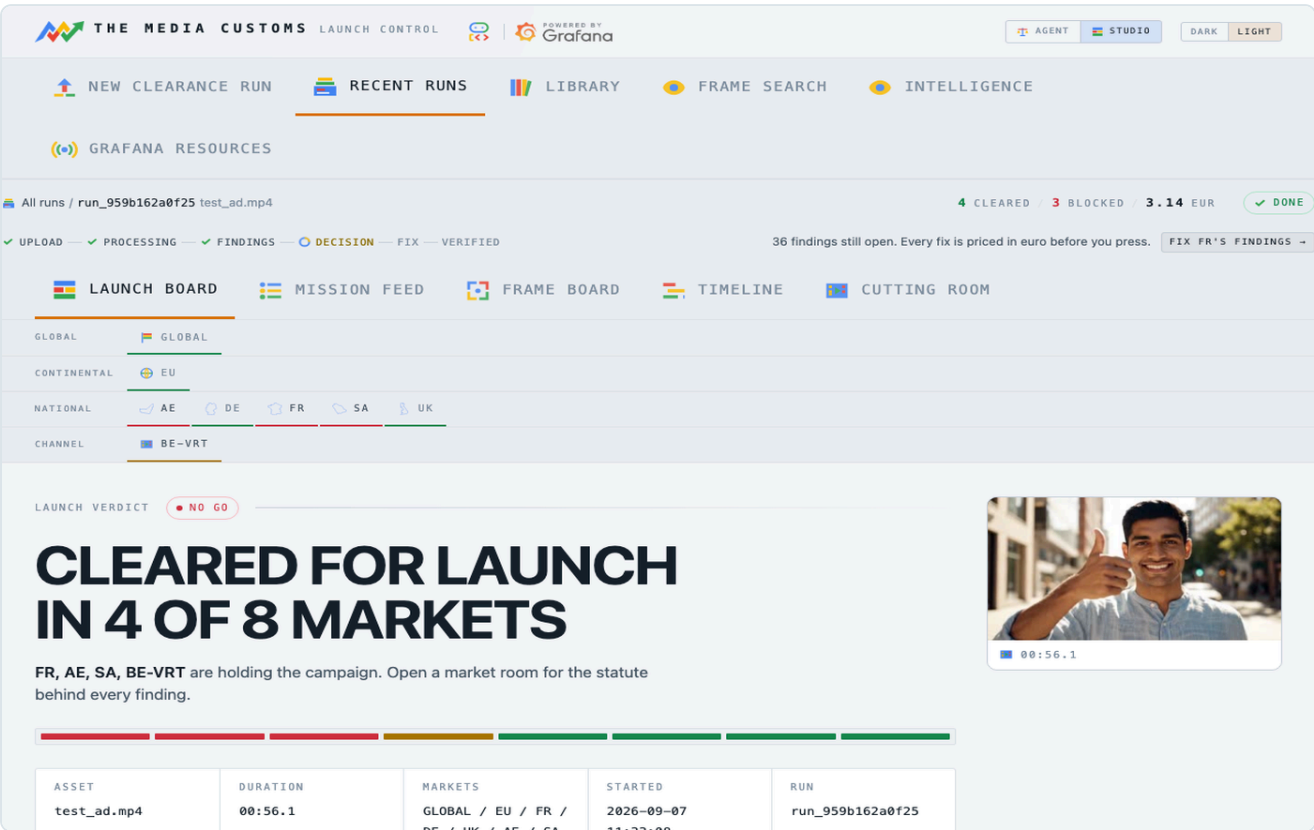
What to look for

LINE	WHAT IT TELLS YOU
judge → FR (6 candidates)	How many observations that market had to rule on.
citation check → FR FR-ALC-01	A citation being resolved against a live source.
concept scope, running anyway	The planner was warned a patch cannot reach this one. Chapter 11.
craft gate passed: 49.7 dB outside the edit	How little of the frame outside the edit box moved.
verify → re-observing 1 touched shot	The verifier looking at the new footage.

The feed is written to the run and survives it, so a finished run shows the same lines in the same order. That is what makes it evidence rather than reassurance while you wait.

The verdict

One tile per market, and the instrument panel the crew built underneath it.



Cleared, at risk or blocked, with the worst finding on the tile.

Reading a tile

STATE	MEANS	DO
Cleared	No finding in this market blocks.	Take the master and the certificate.
At risk	Findings exist but none is at the blocking line, or a citation could not be resolved so its severity is capped.	Read them. This is a judgement call, not a machine one.
Blocked	At least one open, sourced finding is at or over the line.	Open the market room and fix or accept.

Under the tiles is the crew's own lanes dashboard, live from Grafana and clickable. It is not a picture of the run: it is the run's data, drawn by Grafana out of the stores the crew wrote during the run.

An unresolved citation is not a small thing

A finding whose citation will not resolve gets capped severity and can never trigger a fix. That turns a hallucinated rule into a visibly weaker finding rather than a wrong edit.

What it saw

Every scene the crew looked at, and the sentences it wrote before any market saw them.

THE MEDIA CUSTOMS LAUNCH CONTROL POWERED BY Grafana

AGENT STUDIO DARK LIGHT

NEW CLEARANCE RUN RECENT RUNS LIBRARY FRAME SEARCH INTELLIGENCE

GRAFANA RESOURCES

All runs / run_959b162a0f25 test_ad.mp4 4 CLEARED / 3 BLOCKED 3.14 EUR DONE

UPLOAD PROCESSING FINDINGS DECISION FIX VERIFIED 36 findings still open. Every fix is priced in euro before you press. FIX FR'S FINDINGS

LAUNCH BOARD MISSION FEED FRAME BOARD TIMELINE CUTTING ROOM

GLOBAL GLOBAL

CONTINENTAL EU

NATIONAL AE DE FR SA UK

CHANNEL BE-VRT

FRAME BOARD

Frame board

All 11 scenes

00:00.0 00:07.0 00:14.0 00:21.0 00:28.0 00:30.8 00:31.3 00:31.9 00:35.1 00:42.1

SHOT 0 00:00.0 to 00:07.0 6 OBSERVATIONS 6 FINDINGS · AE BE-VRT FR SA

ALCOHOL TOBACCO DRUGS 00:00.0 CONFIDENCE 0.98

A wine bottle with an unbranded label sits on the table between the two women.

FR FR-ALC-01 LEGAL 95 The depiction of a wine bottle constitutes an alcoholic drink depiction restricted under Loi Evvin, Code de la sante publique art. L3323-2.

SA SA-ALC-01 LEGAL 95 The depiction of a wine bottle directly violates the prohibition on alcohol depiction under GAM Media Content Standards. Advertising content in Saudi Arabia strictly forbids depicting, promoting, or consuming alcoholic beverages.

at 00:00.0

SHOW WHAT WAS SPOTTED

How each scene opens and closes, with its neutral observations.

This is the fact sheet all the adjudicators argued about. If a finding looks wrong, come here first and decide which leg of the join failed:

- **The observation is wrong.** The analyst saw something that is not there, or missed something that is. That is a vision problem, and re-running the run is the answer.
- **The observation is right and the rule is wrong.** The market pack needs editing. That is a YAML file, not code. Chapter 23.

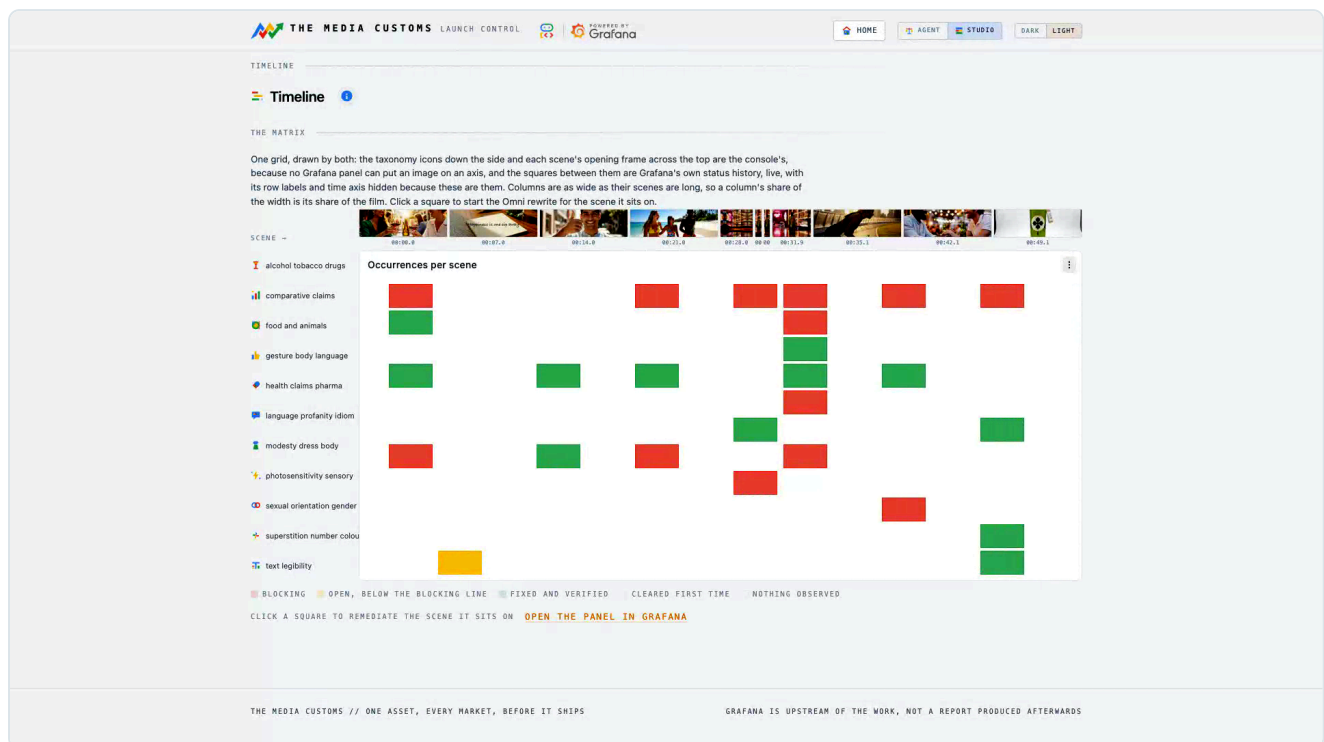
Separating those two questions is the whole reason the analyst is forbidden an opinion.

Every caption here is also a log line

Which is why frame search can answer "which frames show wine" across every run this instance has ever done, without anyone having anticipated the question. Chapter 18.

Where the trouble is

One grid, drawn by two systems, that turns a list of findings into a shape.



Categories down the side, scenes across the top, Grafana in between.

The taxonomy icons and the scene thumbnails on the axes are the console's, because no Grafana panel can put an image on an axis. The squares between them are Grafana's own status history panel, live. A column is as wide as its scene is long, so a column's share of the width is its share of the film.

What you learn here that a list will not tell you

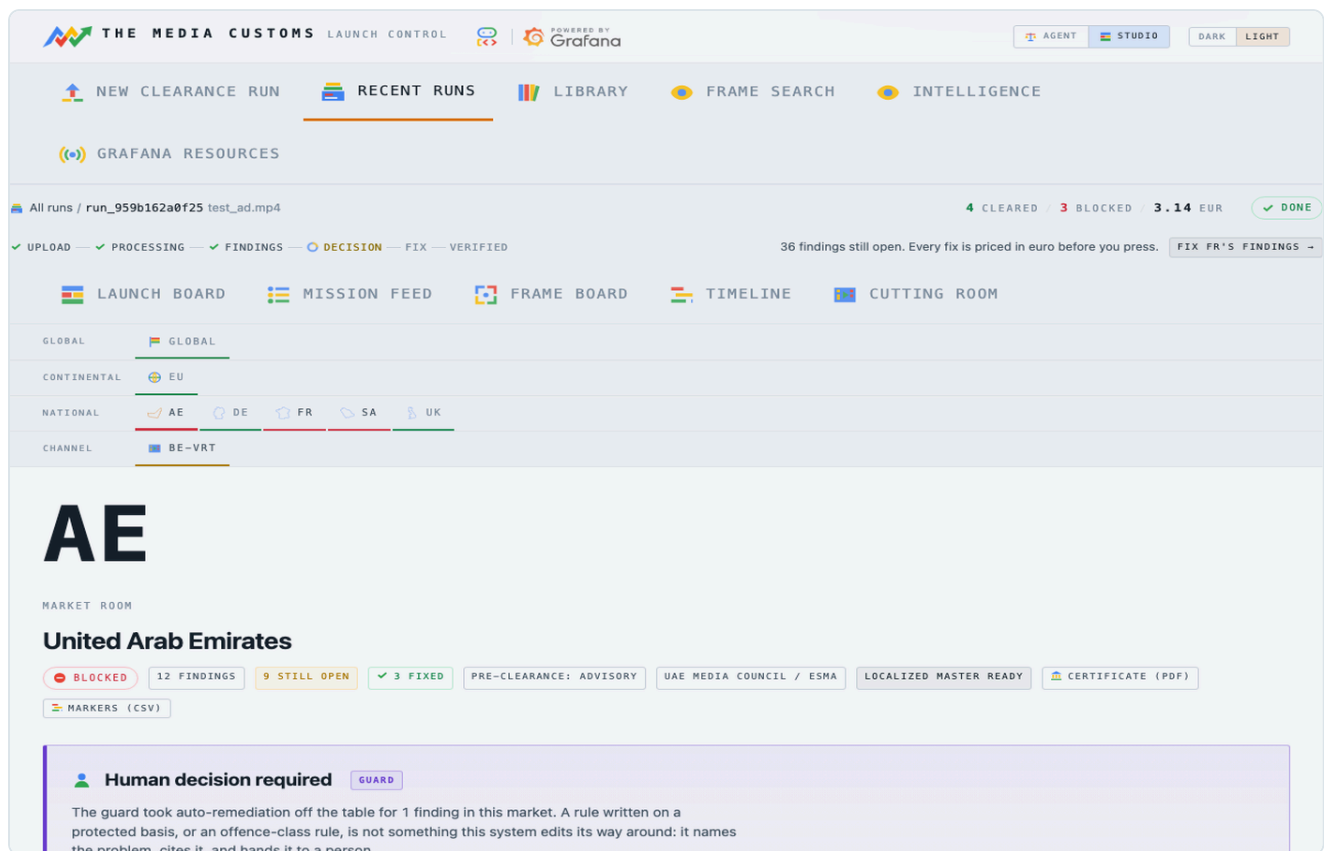
- Whether the trouble is one bad scene or spread through the film.
- Whether one category is doing all the damage.
- Which scene to fix first for the most markets at once.

Clicking a square spends money

A square is a data link. Clicking it resolves the open finding under it and starts a priced fix on that scene. It is the fastest route from "there is the problem" to "fix it", which also means it is the fastest route to a charge. The price is shown before it runs.

One market in full

The market room is where a finding stops being a colour and becomes an argument.



One panel per scene, every finding written against it.

What a finding carries

FIELD	MEANS
Rule id	The rule in the market pack, for example <code>FR-ALC-01</code> .
Class	legal, policy or offence. Legal blocks; offence is never auto-edited.
Severity	0 to 100. Seventy is where a finding starts to block a market.
Window	The seconds it covers, which is what a fix will touch.
Evidence	The triggering frame, with the analyst's own sentence.
Statute	The regulator's own words, and the citation resolved at judging time.
Sourced	Whether that citation resolved. Unsourced findings cannot trigger a fix.

Two takeaways sit at the top of the room: the clearance certificate as a PDF, and the marker files, both per market and both described in chapter 21. When three jurisdictions object to the same two seconds, that is one shot with three objections rather than three problems, and one edit answers all of them.

Fixing a shot

The fix panel asks two questions: what should change, and how it should be done.

Remediate FR-ALC-01

The element runs through the whole scene, but the idea does not depend on it: put a permitted element in its place and the commercial still works. Regenerating the scene is the method most likely to hold, and it is priced accordingly; a patch is still worth trying on a shot that barely moves.

1 WHAT SHOULD CHANGE

- ☒ Replace with water carafe
- ☐ Replace with olive oil bottle
- ☐ Remove bottle from table

2 HOW TO DO IT

⚠ The element runs through the whole scene, so a patch cannot reach it: the scene has to be regenerated with the permitted element in its place.

Method	Cost (EUR)	Fit Rating
☐ Rewrite with Omni any span up to 10 seconds where the footage should stay the brand's own instead of being regenerated	0.70	POOR FIT HERE (7.0s patched · high)
☒ Regenerate with Veo genuine 3D motion, where a patch cannot hold	3.68	8s generated · high
☐ Patch one frame a locked-off shot, where nothing moves	0.04	7.0s patched · low
☐ Propagate the change any shot where the thing to change holds still in frame	0.04	7.0s patched · low
☐ Repaint every frame a target that deforms or is occluded, where one edit cannot be propagated: a bottle travelling to a mouth, a garment on a moving body	3.36	7.0s patched · high

GENERATION BUDGET 44.59 OF 45.00 EUR LEFT TODAY **RUN THE FIX**

SHOT 1 00:07.0 to 00:14.0 1 FINDING · 1 OPEN

SHOT 3 00:21.0 to 00:28.0 1 FINDING · 1 OPEN

SHOT 4 00:28.0 to 00:30.8 1 FINDING · 1 OPEN

SHOT 5 00:30.8 to 00:31.3 1 FINDING

What should change on the left, how to do it on the right, priced.

The left column offers replacements the market allows. The right column offers the five methods, each with its price in euro and a note where it is a poor fit for this particular shot. Nothing is spent until you press.

Read the warning above the methods

When the panel says the element runs through the whole scene, a patch cannot reach it. That sentence is not decoration: it is the scope classifier, and ignoring it is how you pay for an edit that the verifier will reject. Chapter 12 has a worked example.

The five methods

Each one disturbs a different amount of footage, and costs accordingly.

METHOD	WHAT IT DOES	RIGHT WHEN	PRICE
Patch one frame overlay	One image edit of one keyframe, held over the span.	A locked-off shot where nothing moves.	0.04 EUR
Propagate the change track	The same edit, its lighting divided out and multiplied into every live frame.	The thing to change holds still in frame while the shot moves.	0.04 EUR
Repaint every frame per_frame	A repaint of each frame in the span.	A target that deforms or is occluded, where one edit cannot be propagated.	0.04 EUR per frame
Rewrite with Omni omni	Gemini Omni edits the moving footage directly, video to video.	The element runs through the scene, and the footage is yours.	0.10 EUR per second
Regenerate with Veo bridge	Both ends of the span are edited, and Veo 3.1 generates the motion between them.	Genuine 3D motion, where a patch cannot hold.	1.88 to 3.68 EUR

How the automatic choice is made

When an alert triggers a fix rather than a person, the planner picks by the observation's own dimension, and it is a pure function with no model call: relettering for on-screen text, prop_swap for an object, revoice for a spoken line. It escalates to Omni when the scope classifier says the element runs through the whole scene, or when the shot moves too much for a patch to hold.

Veo is never chosen automatically

It regenerates pixels and costs real money, so a bridge only ever runs because an operator picked it, or clicked a data link on a Grafana panel, and the day's budget allowed it.

What each method disturbs, measured

Collateral drift is masked PSNR inside the span, and it is written as a metric on every change. A patch measures about 44 dB. Omni measures between 13 and 28 dB, because Omni re-renders the shot. "Only what you name changes" is the instruction, not the pixels. If preserving the untouched parts of a frame matters more than fixing it convincingly, prefer a patch.

After you press

Three things stand between a generated edit and your master.

1. The craft gate

Before anything is accepted, the staged file is measured: length, resolution, soundtrack, and how much of the frame outside the edit box moved. A file that lost frames, lost resolution, lost its audio or disturbed the rest of the picture is discarded, the master is left untouched, and the finding goes back to open.

2. The verifier

The gate cannot tell you whether the edit is any good, only whether it broke something. So the verifier re-runs the **real analyst pass** over the changed shots and asks the same instrument that found the problem whether it still sees it. It then answers the second half: did anything new break? An edit that removes a bottle can also remove the finding next to it, or introduce one.

3. Grafana closing its own alert

When the finding is verified gone, its blocking metric drops to zero and the alert rule that fired resolves itself. Nothing tells Grafana the fix worked. The number it was watching simply stops being true.

A worked example of a bad fix, and why it was caught

A real change on a real run: asked to "replace each cigarette, cigar or tobacco pack", Omni turned a cigar into a cigarette. One banned item for another, which is the smallest edit that satisfies the sentence and no fix at all. The craft gate passed it, because almost nothing outside the edit had moved, which is exactly what the gate measures. The verifier then re-observed the shot, the tobacco rule fired again, and the finding went back to open. Nothing was signed off. The instruction was what was wrong, and it now says out loud that the replacement must not itself be alcohol, tobacco, a smoking device or a drug in any form.

When the fix is wrong

The verifier can only ask its own question. A fix can pass it and still be wrong to a person.

An Omni rewrite once took a tobacco plug out of a hand and left the character holding something rifle-shaped. It cleared its rule, because the rule was about tobacco. To a human it was obviously not shippable.

So every change on the **my edits** screen carries a **re-edit this** control. You say what is wrong with it in your own words, and your words become the instruction for the redo.

How to write a good re-edit reason

INSTEAD OF	WRITE
"this is bad"	"he should not have a gun in his hands afterwards"
"wrong colour"	"the replacement bottle should be clear glass, not green"
"still visible"	"the ashtray on the table is still there in the second half"

Name the object and where it is. The instruction is handed to the model as you wrote it.

A patch is redone with Omni

Words only reach a model that is given the span. A patch method edits a still, so it cannot act on "afterwards" or "in the second half". A re-edit of a patch is therefore promoted to Omni, which changes the price. It is logged on the run either way, whatever comes back.

When it will not edit

When a rule is written on a protected characteristic, the honest answer is not an edit.

Two rules in the corpus carry a protected basis. A finding matching either is marked **remediation blocked** with the verbatim reason *rule basis targets a protected characteristic; human decision required*, and the console shows the statute beside it.

What the Guard reads

- The pack rule matched by its rule id.
- The finding's own class.

That is all. It never reads the rationale, the severity or any other model-authored field, and it never calls a model. It is a pure function, which is what makes it **un-promptable**: a crafted finding cannot argue its way past it. The refusal is enforced a second time at the point of action, before a single frame is touched.

The three ways out, and they are all a person's

CHOICE	WHAT IT MEANS
Approve a manual edit	An editor fixes it in the suite. Take the markers.
Send to legal review	The statute goes with it. Recorded on the run.
Accept the risk	Air it as it is, knowingly. Also recorded, and printed on the certificate.

Which rules carry it

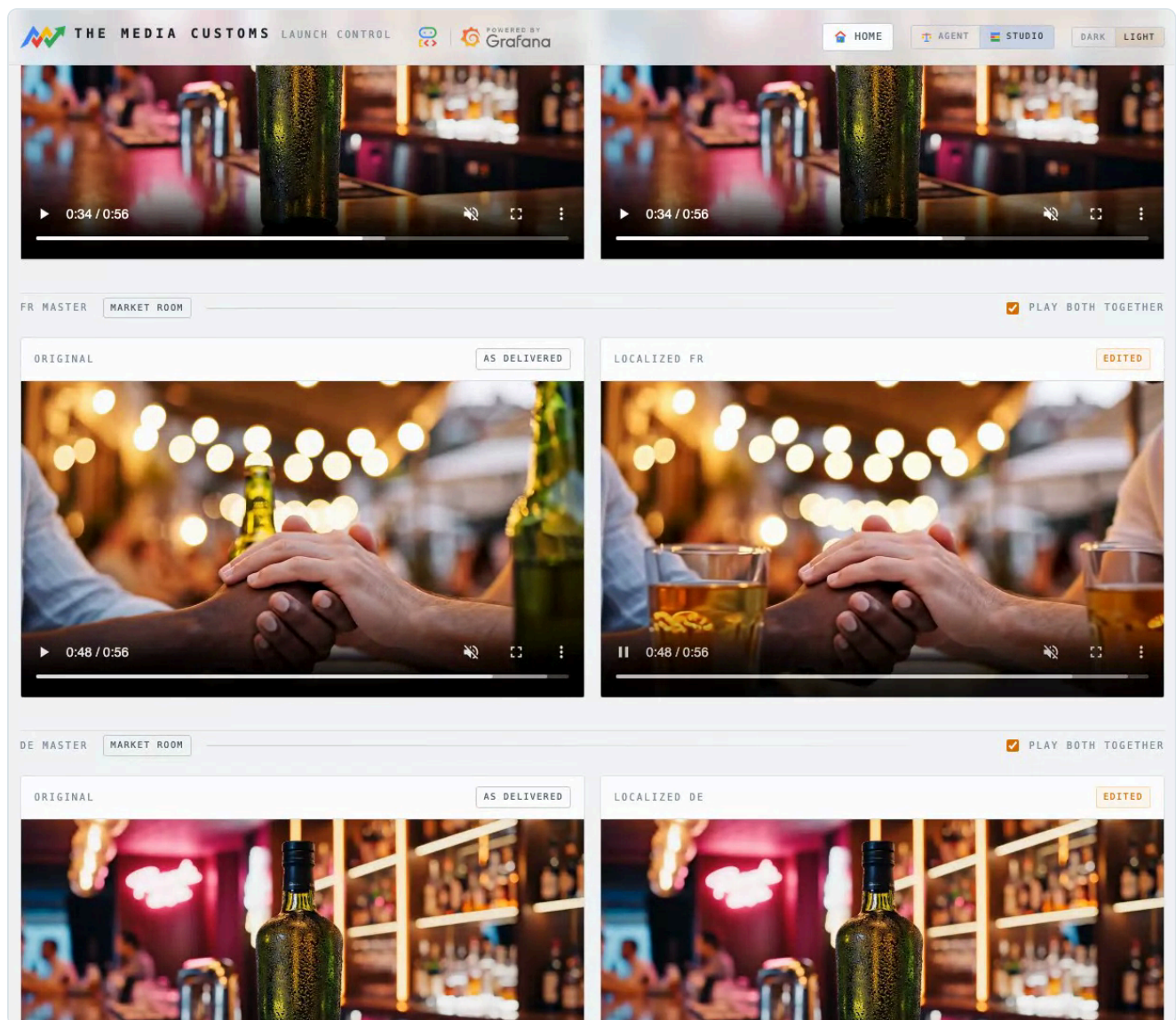
A pack marks a rule with `protected_basis`. Two rules in the corpus carry it today, both about who may be shown together, and both national law rather than a broadcaster's preference. Any pack can set it, and any rule that sets it gets this treatment.

Why this is a feature and not a gap

Guardrails belong in rule layers, not prompts. Anything a prompt grants, a prompt can take away. A system that quietly edited people out of an ad to satisfy a rule about who may be shown together would be doing something no operator asked for, in your name.

Checking the result

The original and the localized master, playing in lockstep on the second that changed.



One pair per market. Both players open on the changed second.

One edited master per market, written by the remediator and confirmed by the verifier. The pairs are stacked by market and each opens on the second that changed rather than at zero, so you are looking at the edit within a second of arriving.

Generated content

A second screen shows everything a model produced for the run: the stills a patch wrote, the seconds Omni or Veo rendered, and for a bridge the two anchor frames it was handed. Those two frames are the whole brief, so they are the only way to tell whether Veo invented something.

What to check: is the violating thing gone, did anything else change, does the edit hold for the whole span, and does the cut still work?

Across every run

My edits is the cross-run cutting room: what has this thing actually changed?

Every edit the instance has made, newest first, with the frame before and the frame after side by side and the rule that asked for it named. One card per scene, not per market, and the card names the market whose objection paid for the render. Picture edits and sound edits are two sides of a toggle, because one you watch and the other you listen to.

Removing an edit

This is the one control in the console that destroys evidence, so it asks for a word and says exactly what it takes: the change record, both frames and any generated clip. There is no undo, and they are the only copy.

1. Open **remove** on the card.
2. Type the word and press **remove this edit**.
3. The card disappears at once. The deletion continues behind it.

What a card shows you

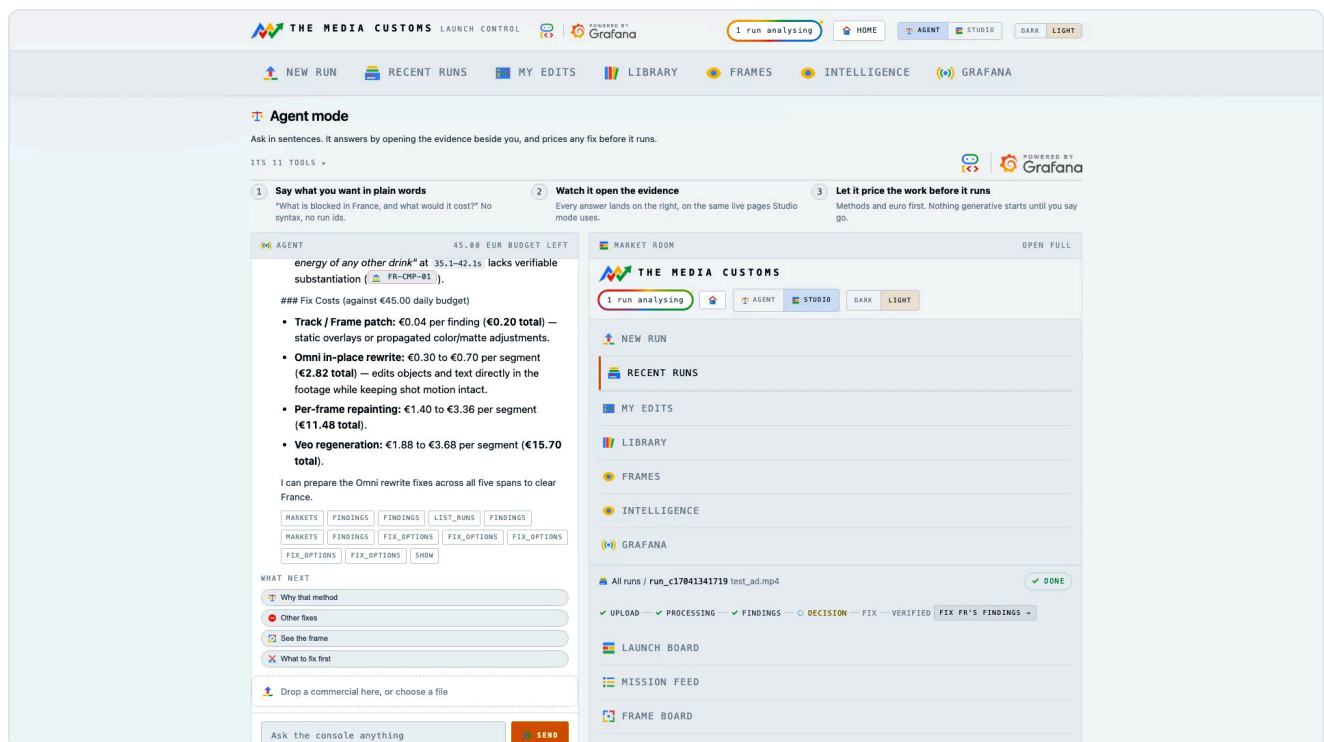
The frame before and the frame after, the market that paid for the render, every market that objected to those seconds, the method that made it, and the change id. Hovering plays both sides together, so a fix that only works in one frame gives itself away immediately.

The card going does not mean it is gone yet

Deleting a row, two stills and a rendered clip takes a moment, and a card that sat there until it finished read as a button that did nothing. So the card is hidden the instant you press, and the work carries on behind it. **If the server refuses**, because the word was wrong or the change was already deleted, the card comes back with the reason under the form and nothing was removed. A message you can read is the difference between an optimistic interface and a lying one.

Asking in sentences

A second agent, with ten tools and the whole console as its canvas.



It answers on the left and opens the evidence on the right.

Agent mode answers by opening the view that proves the answer, and it prices any fix before it runs. It reads every run in the instance, not just the one you are looking at.

Questions that work well

ASK	YOU GET
Why is France blocked, and what would a fix cost?	The rules, the seconds, the statutes, and a price per method.
Show me the frames FR-ALC-01 fired on	The frames themselves, opened beside the answer.
What should I fix first?	The shot that closes the most markets for the least money.
Which markets need a human?	The Guard's refusals, with their statutes.
Build me a dashboard for this run	A real Grafana dashboard, from the run's own labels.

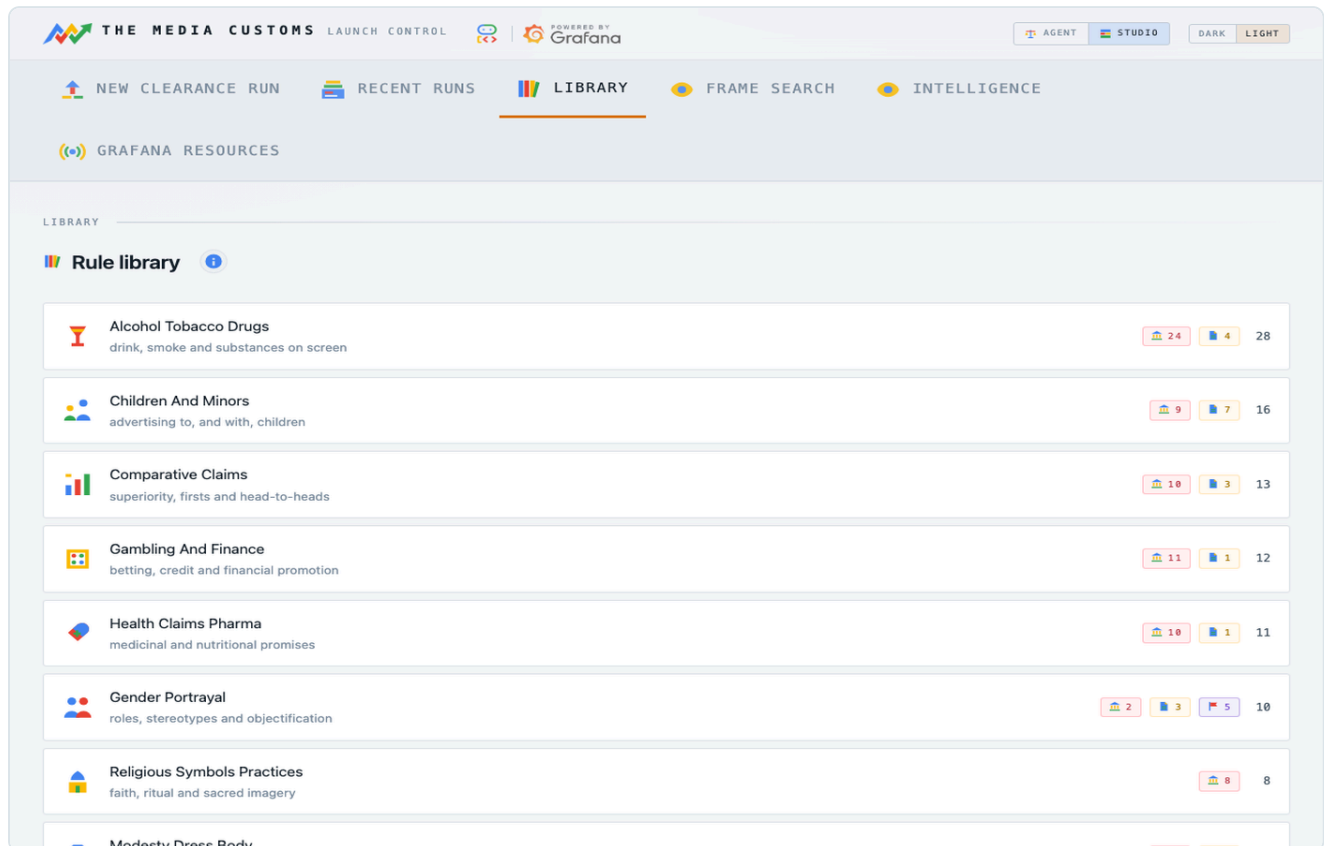
It spends, and it says so first

Asking costs a model call. Running a fix costs what the fix costs. The agent quotes before it acts and refuses to spend past the day's ceiling, but it is not a free text box.

Looking things up

Two reference screens: what the rules say, and where a thing appears.

The rule library



Every rule, filed under the observation that can trigger it.

128 rules across 21 packs, each showing the market that holds it, the statute text, whether it blocks or advises, and which of the eighteen dimensions can trigger it. This is the place to answer "what would this market object to" before you have run anything.

Frame search

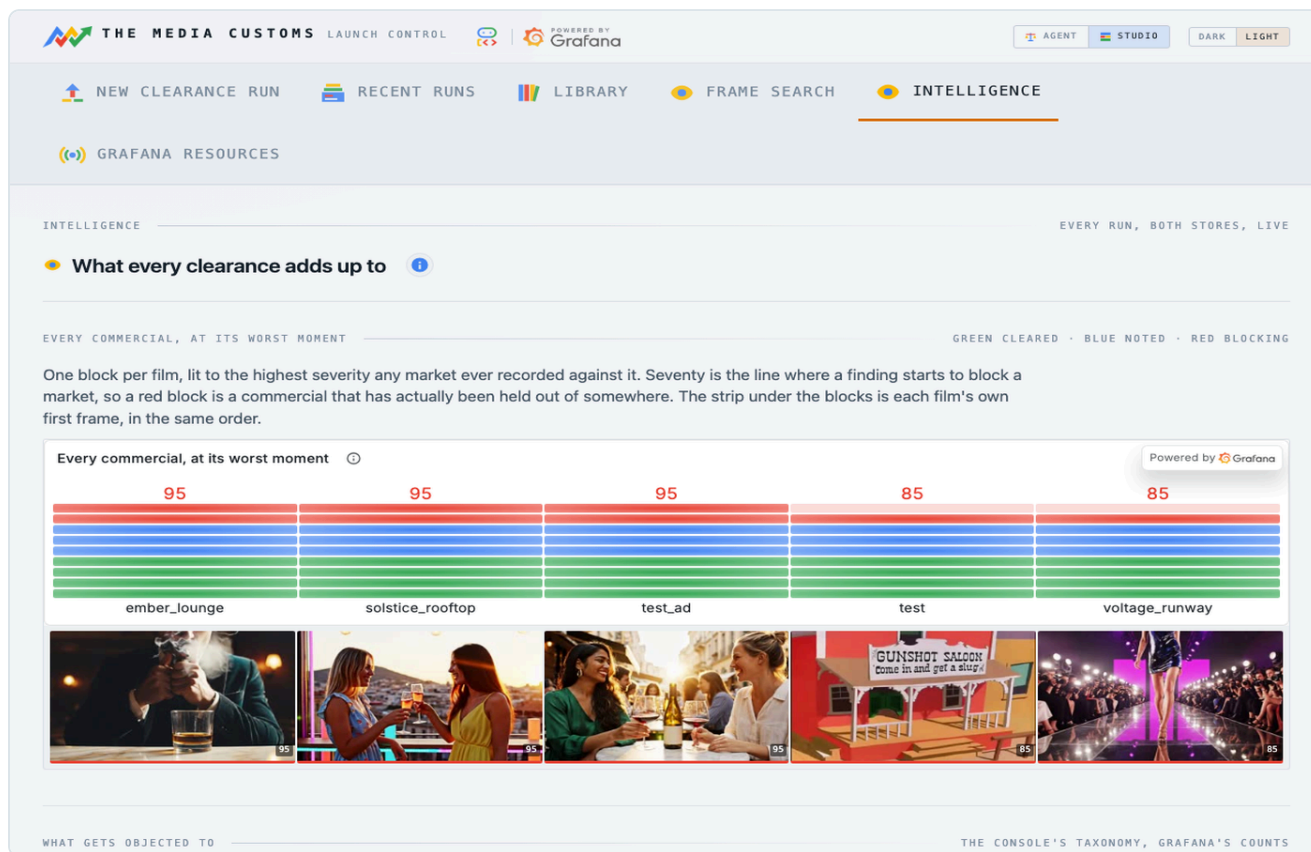
Every caption the analyst ever wrote is a Loki line, so "which frames show wine" is a question rather than a feature somebody anticipated. The match is semantic: the question and the candidate captions go to Gemini together, so *bunnies* finds *an animated rabbit character* without anybody guessing which word a vision model chose months ago. Each result says, in the model's own words, why it is there.

Two modes

Semantic is the default and costs a model call. `mode=literal` is a regex over the caption alone and costs nothing: use it when you mean an exact string, a rule id or a brand name.

Reading across runs

Every other screen answers a question about one commercial. This one reads across all of them.



Seventeen panels and eight kinds of chart, from the same two stores.

Which subject your creative keeps tripping over, which jurisdictions are actually hard, the dimension against market matrix, the severity distribution, worst severity per market against the line where a finding starts to block, market status over time, and the raw finding stream underneath it all.

The division of labour, and why it looks like this

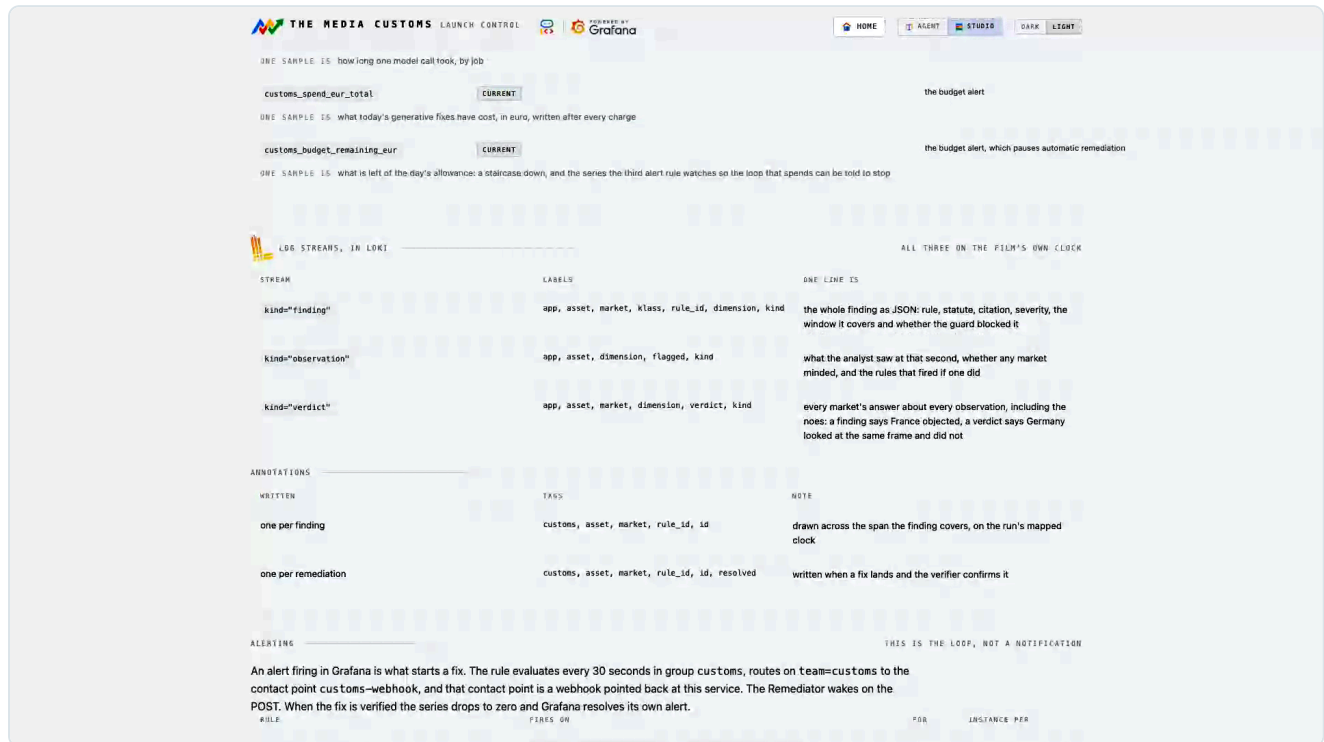
Grafana charts the data because it holds it. The console draws the axis because Grafana has never heard of an eighteen-part taxonomy or a broadcaster channel's mark. The hero is one block per commercial, lit to the highest severity any market ever recorded against it, with that film's own first frame underneath it in the same order.

This is where a pattern shows up before it is expensive

One blocked ad is a problem. The same dimension blocking four ads in a row is a brief that needs changing, and it is only visible from here.

What it built in Grafana

Not a report produced afterwards. The instrument panel is built during the run, by an agent, and it is what triggers the work.



The whole surface on one page, read from the definitions the agent provisions from.

WHAT	HOW MUCH
Dashboards, built at runtime	9 boards, 40 panels
Metric series to Mimir over OTLP	9 series
Log streams to Loki	3 kinds
Alert rules	3
Write operations, MCP where a tool exists	8

The crew provisions its own folder, dashboards and alert rules through the official mcp-grafana server. Where that release has no write tool for an operation, it goes over the provisioning API instead and the inventory says so.

Two clocks, and this is the thing people get wrong

One series is written on the **film's own clock**: video second n is pushed at wall second t_0 plus n , so a panel's x axis reads as the timecode because it is the timecode. The others are stamped at the **real clock**, because an alert rule's evaluation window has to mean what it says. Point the stock dashboards at "last 6 hours" and they will say No data. Pin the range to the run.

The three alert rules

Two ask Grafana to wake the crew. The third asks it to stop.

RULE	FIRES WHEN	WHAT HAPPENS
customs-blocking	customs_blocking crosses 70 for an asset, market and rule	The webhook wakes the remediator on that finding.
customs-market-down	A market's status goes to at risk or blocked	Recorded, and visible on the overview.
customs-budget-low	customs_budget_remaining_eur falls to the floor	Automatic remediation pauses for the rest of the UTC day.

The rules evaluate every thirty seconds. The contact point is a route on this service, and the webhook reads **labels only**: it never trusts an annotation or a message body to tell it which finding to spend money on.

The series worth knowing

SERIES	CLOCK	ONE SAMPLE IS
customs_risk	mapped	the worst severity in force at video second n
customs_market_status	current	0 cleared, 1 at risk, 2 blocked
customs_blocking	current	the severity of one open, sourced, blocking finding
customs_stage_error	current	how many times one stage failed on this asset
customs_collateral_drift	current	PSNR in dB inside the span of one edit
customs_spend_eur_total	current	what has been charged today

Why the panels in the console are sometimes pictures

Grafana Cloud answers every embed attempt with an enforcing frame-ancestors policy, so it cannot be put in an iframe. Where the companion viewer is deployed, the console frames live panels reading the same stores. Where it is not, it renders server-side PNGs instead.

What you get out

Four artefacts. Two are for machines, two are for people.

The clearance certificate

A PDF per market, and deliberately not a summary. Every finding the market raised is on it, worst first, with the rule id, the class, the severity, the seconds it covers, the statute in the regulator's own words, and the citation the adjudicator retrieved at judging time. A fixed finding carries the method, the change id and the verifier's own sentence. A finding the Guard refused says **human decision required** with the statute behind it, and the decision a person made. An unsourced rule says that its severity was capped, and why.

It is the document an archive needs three years later when somebody asks why this cut aired in Riyadh.

The localized master

One edited file per market, with only the failing seconds touched, checked by the craft gate and signed off by the verifier. Downloadable from the cutting room.

Timeline markers

Two formats, because the two suites disagree about which they trust: a CSV with the columns both of them look for, and CMX3600, the oldest interchange there is and the one Resolve imports most reliably. Both carry the frame rate they were written at, read as the rational the probe reports rather than as a float, because 23.976 written as 24 drifts a frame a second and looks perfectly plausible while it does it.

COLOUR	MEANS
Red	A legal finding at or over the blocking line.
Green	The verifier already signed a fix off here.
Yellow	Everything else still open.

A finding the Guard refused carries **HUMAN DECISION REQUIRED** in its own note, so the editor reads why while they work.

The money

The loop that can spend is the loop that needs a brake.

Judging is cheap and generating is not. Every method shows its euro cost in the picker before you press, and the agent quotes a fix before it starts one. Every charge is written to Mimir on the real clock as it happens, so spend is a series you can chart rather than a number somebody reconciles later.

The daily ceiling

CONTROL	EFFECT
Generation budget	A euro ceiling for the whole system, per UTC day.
Visitor cap	A much smaller ceiling for anyone through the visitor door.
Budget alert	Fires above the price of the most expensive single fix, and pauses automatic remediation.

What the pause does and does not do

- Findings still block. Nothing is silently cleared.
- Alerts still arrive and are recorded in the feed **with the reason**.
- A person at the console can still spend what is left, one fix at a time.

Why the brake is on that path only

The pause is on the path nobody is watching, which is the only one that needed one. An operator pressing a button has already seen the price. An alert firing at three in the morning has not.

A rough sense of scale

A judging run across eight markets costs cents. A patch costs about four cents. An Omni rewrite of a six second span costs about sixty cents. A Veo bridge costs between one and four euro. A day's budget therefore buys somewhere between a dozen and two dozen bridges, or hundreds of patches.

Adding a market

A market is a YAML file, not code. The pack format is the extension point.

A pack names its market, its parent on the ladder, and its rules. Each rule names the dimension that can trigger it, the class, the severity it carries, the statute in the regulator's own words, and a citation reference that has to resolve.

```
market: FR
name: France
parent: EU
rules:
  - id: FR-ALC-01
    dimension: alcohol_tobacco_drugs
    class: legal
    severity: 95
    trigger: alcohol is visible or being consumed on screen
    statute: >
      Loi Evin, Code de la sante publique art. L3323-2: television
      advertising for alcoholic beverages is prohibited.
    citation_ref: legifrance.gouv.fr LEGIARTI000006688087
```

The four things a good rule has

1. **One dimension.** If it needs two, it is two rules.
2. **A trigger written as what the analyst would see**, not as what the law forbids. The analyst never reads your rule; the adjudicator joins them.
3. **A real statute**, quoted rather than paraphrased.
4. **A citation that resolves.** If it does not, findings on this rule get capped severity and can never trigger a fix, which is the system protecting you from your own typo.

Inheritance is free

Declare a parent and you inherit everything above you. A broadcaster pack usually only needs its own acceptance rules: its country and the continental and global rungs come with it.

protected_basis is not a severity dial

Setting it marks a rule as written on a protected characteristic, which routes every finding on it to a person and blocks automatic editing. Use it for what it means, not to make a rule feel important.

Running it yourself

Install, configure, provision, deploy.

Prerequisites

- ffmpeg on the PATH, and Python 3.12.
- A Google Cloud project with Vertex AI enabled.
- A Grafana Cloud stack, with Mimir and Loki.

Install and run

```
# 1. Environment
python3.12 -m venv .venv && .venv/bin/pip install -r requirements.txt
cp .env.example .env          # then fill in the Google Cloud and Grafana values

# 2. Provision the Grafana surface: dashboards, all three alert rules,
#    the webhook contact point, the share links. Idempotent.
.venv/bin/python scripts/provision_grafana.py

# 3. Clear an ad from the terminal
.venv/bin/python scripts/run_pipeline.py docs/samples/test_ad.mp4 --markets FR,EU,AE

# 4. Or run the console and use a browser
.venv/bin/uvicorn customs.app:app --reload    # http://127.0.0.1:8000
```

Configuration worth knowing

VARIABLE	WHAT IT SETS
GOOGLE_CLOUD_PROJECT	The Vertex project every model call bills to.
GRAFANA_URL, GRAFANA_SA_TOKEN	The stack the crew provisions into.
LOKI_PUSH_URL, OTLP_URL	Where lines and metrics go.
WEBHOOK_TOKEN	The key on the alert contact point. Without it the webhook refuses.
JUDGE_PASSWORD, VISITOR_PASSWORD	The two doors.
EDITS_PASSWORD	The word that removes a change record.
VEO_MODEL, OMNI_MODEL	Pinned model ids, re-probed on a cold start.

Deploying it

Two scripts: the service, and the viewer that lets its panels be framed.

```
scripts/deploy.sh          # the service, secrets, IAM and the Grafana wiring
scripts/deploy_viewer.sh   # the embeddable Grafana viewer beside it
```

The deploy script is idempotent and safe to re-run. It enables the APIs, mints the secrets, sets the runtime service account, builds the container, deploys the revision, and re-wires the Grafana contact point to the new URL.

Two things it does to protect you

- **It refuses while work is in flight.** A deploy replaces the running container, so a clearance or a fix mid-render would be discarded. It asks the live service first and stops if the answer is yes.
- **It re-points the webhook.** The alert contact point has to name the service that is actually running, or a blocking finding would fire an alert into nothing.

Check the revision, not the exit code

After it finishes, confirm the serving revision actually changed. A build can succeed while the service keeps serving the old one, and the script's exit code will not tell you.

Where state lives

Runs survive a deploy: the store is mirrored to object storage and restored on boot. A run killed mid-flight by a deploy is still lost, which is the other reason the script refuses.

When something goes wrong

The failures this system actually has, and what each one means.

WHAT YOU SEE	WHAT IT IS	WHAT TO DO
A dashboard says No data	The time range is wall clock, but the run's risk series is on the film's clock.	Pin the range to the run, or open the panel from the console.
Omni refused the input	The footage contains recognisable third-party content.	Nothing was charged, and a retry cannot help. Use a patch method, or your own footage.
Omni refused the output	The instruction asked for something it will not render, often about a photorealistic person.	Rewrite the instruction. The footage was fine; the ask was not.
Vevo refused the frames	Its filter read the shot as depicting a public figure.	Never charged. The footage is the refusal.
A fix passed and the finding reopened	The verifier saw the violation survive, or something new break.	Read the new finding's words. Often the method could not reach it. Chapter 11.
A fix looks wrong but cleared	The rule was satisfied and a person still would not ship it.	Use re-edit and say what is wrong. Chapter 13.
Everything is slow, then a stage errors	One instance, one writer thread. Two concurrent multi-market clearances are too much.	Run them one at a time.
The deploy refuses	The live service reports work in flight.	Wait. Forcing it discards a real clearance or a fix mid-render.
A fix seems stuck for a long time	Image generation quota is a small number of requests a minute, project wide.	Per-frame methods are slow by arithmetic. Prefer Omni for long spans.

The general rule

Every stage that fails says so in the mission feed and counts itself as a metric. A tool that silently skips a shot is worse than one that admits it, so if the feed is quiet the work really is still running.

Reference

Routes, stores and streams, for looking something up quickly.

Console routes

ROUTE	SCREEN	GATED
/	Front door	no
/runs	Archive	no
/new	New clearance	spends
/runs/{id}	Launch board	no
/runs/{id}/mission	Mission feed	no
/runs/{id}/frames	Frame board	no
/runs/{id}/timeline	Timeline	no
/runs/{id}/markets/{m}	Market room	no
/runs/{id}/cutting	Cutting room	no
/runs/{id}/generated	Generated content	no
/edits	My edits	removal spends
/agent	Agent mode	spends
/library	Rule library	no
/search	Frame search	no
/insight	Intelligence	no
/grafana	Grafana resources	no
/webhook/alert	The alert contact point	keyed

Reference: the stores

What the crew writes, and how to read it back.

Log streams in Loki

KIND	ONE LINE IS	LABELS
observation	one keyframe the analyst described	app, asset, dimension, flagged
finding	the join, with the statute in it	app, asset, market, rule_id
verdict	one market's answer	app, asset, market

Per market, without a scan

Because market and rule id are labels rather than fields, one market's findings are a selector: `{app="customs", kind="finding", market="FR"}`.

The two clocks, once more

CLOCK	SERIES	READ BY
Mapped , the film's own	customs_risk	The timeline, the lane strip, the grid.
Current , the wall clock	customs_market_status, customs_blocking, customs_stage_error, customs_spend_eur_total	The overview, and all three alert rules.

Alerting reads only the current-clock series, because an evaluation window has to mean what it says. The timecode axis reads only the mapped one.

If you remember one thing from this manual

Every screen here is a different distance from the same question: what is in this film, and who objects to it? The findings are joins you can take apart, the fixes are priced before they run, and nothing is signed off because a model said so.



THE MEDIA CUSTOMS

One asset, every market, before it ships.



The console, live and open to read:
`customs-app-akap4ao72a-ew.a.run.app`

The code:
`github.com/tdries/devpost-hackathon-agentic-cinema`

Built on Google Cloud and Grafana. Operator manual v1.0.